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CV

Current academic and professional CV covering secure AI, blockchain, distributed systems, energy cybersecurity, publications, patents, and consulting experience. Download the PDF version using the button above.



General Information

Full Name	Dr. Samuel Karumba
Location	Sydney, Australia
Languages	English, Swahili
Citizenship	Kenya
Australian Visa Status	Temporary Graduate Visa (Subclass 485, Post-Higher Education Work stream) - Full Work Rights
Email	karumba66@gmail.com
ORCID	0000-0002-7181-9799
Google Scholar	XL1vtsgAAAAJ

Bio Statement

Dr. Samuel Karumba designs secure, privacy-preserving distributed systems across blockchain, AI, energy cybersecurity, and digital trust infrastructure. His work spans CSIRO Data61, UNSW, IBM Research Africa, and independent consulting, with contributions to blockchain interoperability, verifiable AI, carbon-market traceability, consumer energy resource cybersecurity, and patented IoT/blockchain systems. He works across African and Australian research ecosystems on cryptographic protocols, privacy-preserving computation, and resilient decentralized system design.

Education

2019 - 2024	Doctor of Philosophy (PhD) in Computer Science University of New South Wales (UNSW), Sydney, Australia - Thesis: Blockchain-based Decentralized Energy Systems: Advancing Scalability, Security, Privacy, and Interoperability - Advisors: Prof. Salil S. Kanhere (UNSW), Prof. Raja Jurdak (QUT), Dr. Subbu Sethuvenkatraman (CSIRO) - Developed 4 blockchain frameworks (HARB, PlexiChain, CypherChain, BAILIF) adopted for energy security use cases - Published 7 research papers linked to thesis chapters - Influenced energy governance standards in Australia and globally - Awarded UNSW Scientia Scholarship (Top 3% of PhD applicants) - DOI: https://doi.org/10.26190/unsworks/28768
2013 - 2017	Master of Science (MSc) in Mobile Telecommunication and Innovation Strathmore University, Nairobi, Kenya - Dissertation: Smart Education: An Event Framework for Cognitive Blended Learning - Advisors: Dr. Joseph Sevilla, Dr. Vitalis Ozianvi

- Designed and piloted a cognitive learning companion prototype
- Awarded Safaricom Academy Scholarship (Top 25 applicants from Kenya)

2008 - 2012

Bachelor of Science (BSc) in Computer Science



Moi University, Eldoret, Kenya

- Second Class Honours (Upper Division)
- Final year project on secure database synchronization in distributed networks
- Core courses: Software Engineering, Distributed Systems, Database Systems, Computer Security and Cryptography, Algorithms and Data Structures, Artificial Intelligence

Experience

JAN 2025 –
PRESENT

Research & Innovation Consultant



Independent – Secure Decentralized Systems, Sydney, Australia

- Leading design and deployment of blockchain-based data integrity systems and verifiable AI infrastructures for fintech and energy startups across Africa and Australia
- Architecting Hyperledger Fabric networks supporting identity management, permissioned data exchange, and smart contract governance for compliance-aware applications
- Developing modular data pipelines integrating blockchain and analytics for AI explainability and traceability in distributed systems
- Conducting scalability, performance, and cryptographic audits for early-stage blockchain ventures, ensuring resilience and regulatory compliance
- Delivering DevOps pipelines for blockchain platforms, integrating containerized testing, monitoring, and CI/CD automation using Docker, Kubernetes, and GitHub Actions
- Advising DeFi remittance startups on tokenization of fiat into stablecoins, cross-border settlement flows, and controls for on/off-ramp compliance

DEC 2023 – DEC
2024

Postdoctoral Research Engineer



CSIRO Data61, Sydney, Australia

- Led engineering of cybersecurity and blockchain-based assurance frameworks within the CAESER program (Cybersecurity of Consumer Energy Resources)
- Designed permissioned blockchain prototypes for data provenance and device attestation across smart inverters and distributed energy systems
- Integrated AI-driven anomaly detection and risk classification models for grid-edge systems, improving incident detection precision by 30%
- Authored peer-reviewed publications translating research into operational frameworks adopted by Australian utilities and policy bodies

2023

Postdoctoral Researcher



University of New South Wales (UNSW), Sydney, Australia

- Researched blockchain for voluntary carbon markets
- Produced policy recommendations adopted by 3 carbon firms
- Co-designed selective disclosure protocol using zero-knowledge proofs
- Improved digital MRV architecture leading to 20% increase in process transparency

2022 - 2023

Research Assistant

 University of New South Wales (UNSW), Sydney, Australia

- Researched blockchain for voluntary carbon markets
- Produced internal report on selective disclosure and traceability in carbon credit systems
- Collaborated with legal experts to align with international voluntary carbon market standards

2019 - 2023

PhD Researcher

 CSIRO & UNSW, Newcastle & Sydney, Australia

- Developed cryptographic, scalable, and privacy-preserving blockchain solutions for decentralized energy systems
- Published 7 first-author research papers during PhD
- Prototyped and evaluated 4 novel blockchain frameworks addressing scalability, privacy, and interoperability
- Presented research at top-tier conferences including ICBC, ACM e-Energy, and ICTD
- Mentored multiple postgraduate students and contributed to research training workshops
- Developed simulation environments and metrics used by subsequent research teams

2020 - 2023

Lecturer / Guest Instructor

 University of New South Wales (UNSW), Sydney, Australia

- Delivered guest lectures on blockchain scalability and privacy (COMP9321 - Data Services Engineering, Postgraduate)
- Provided lab supervision and tutorials on web application security (COMP6443 - Web Application Security, Undergraduate)
- Supervised 5 research thesis projects (Honours and Masters levels)
- Topics: Privacy-preserving data sharing in smart grids, Secure IoT frameworks, Blockchain-based credential verification, Multi-chain identity authentication
- Outcomes: 3 students published papers in peer-reviewed venues, 2 students secured scholarships for postgraduate research

2015 - 2019

Research Software Engineer

 IBM Research Africa, Nairobi, Kenya

- Built blockchain-enabled mobile, IoT, and AI systems for agriculture, education, and logistics
- Co-invented and granted 3 US patents in blockchain and IoT systems
- Piloted Agri-Wallet impacting 2,000+ smallholder farmers
- Co-developed IBM cognitive learning companion and school analytics platform adopted by county education ministries
- Contributed to early prototyping of TradeLens, IBM's global shipping blockchain platform
- Delivered technical workshops and training to regional government and NGOs

2014 - 2015

Research Intern



IBM Research Africa, Nairobi, Kenya

- Contributed to Cognitive Learning Companion (CLC) project
- Designed iOS-based content delivery platform
- Instrumented mobile phones for multimodal sensor data collection and real-time behavioral feedback
- Co-developed cooperative download architecture for efficient offline content sharing in resource-constrained environments

Research Projects

2019 - 2022

HARB - Hypergraph-Based Adaptive Consortium Blockchain

- Framework for decentralized energy trading using hypergraph theory
- Published in IEEE Internet of Things Journal (Impact Factor: 8.2, Q1)
- Outperformed single-chain and PoA models in simulation

2021 - 2022

PlexiChain - Secure Blockchain-Based Flexibility Aggregator

- Privacy-preserving framework integrating PUFs and NFTs for energy flexibility aggregation
- Published as arXiv preprint (arXiv:2212.09064)
- Formed security backbone for BAILIF cross-chain validation

2022 - 2024

CypherChain - Privacy-Preserving Data Aggregation Framework

- Secure multiparty computation for blockchain-based demand response programs
- Published at IEEE ICBC 2024
- Maintains transactional transparency while preserving anonymity

2022 - 2023

BAILIF - Blockchain Agnostic Interoperability Framework

- Cross-chain interoperability solution for energy systems
- Published at IEEE ICBC 2023
- Enables secure atomic swaps and cross-chain data exchange

NOV 2023 - OCT 2024

CAESER - Cybersecurity of Consumer Energy Resources

- National cybersecurity project for Australian smart grids
- Event-driven resilience, machine learning, and policy-guided cyber defense
- {"Project period"=>"November 2023 - October 2024"}
- Cross-institutional collaboration with CSIRO, Edith Cowan University, and UNSW

2022 - 2023

CCDI - Carbon Certification & Development Initiative

- Blockchain-based framework for voluntary carbon credit verification
- 3 industrial pilot engagements across Southeast Asia
- Improved carbon credit traceability by 20%

2018 - 2019

Agri-Wallet for Smallholder Farmers

- Blockchain-powered payment and booking system for mechanized agricultural services
- Impacted 2,000+ farmers in pilot
- Granted patent (US11497156B2)

2016 - 2018

TradeLens Global Supply Chain Blockchain

- Enterprise blockchain platform for global supply chain digitization

- Contributed to early prototyping and development

Honors and Awards

- 2018

 - UNSW Scientia PhD Scholarship (Top 3% of PhD applicants, ~900 applicants across disciplines)
- 2013

 - Safaricom Academy Scholarship (Top 25 applicants from Kenya, ~200 applicants)
- 2023

 - ACE Summer School Travel Stipend (Monash University / ACE Program)

Publications Summary

TOTAL CITATIONS

340

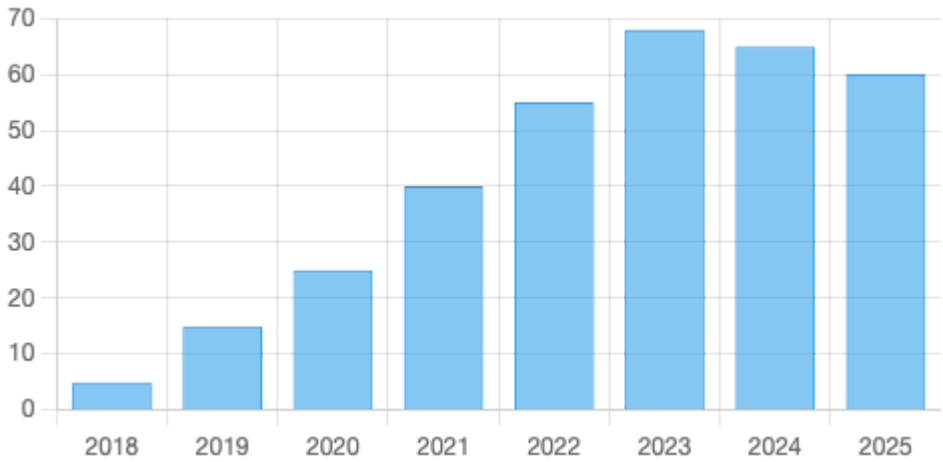
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CITATIONS BY YEAR



Downloadable CV Versions

[General academic and professional CV](#)

[Blockchain and digital trust CV](#)

[Data science, AI, and analytics CV](#)

[Python and software engineering CV](#)

Research Interests

Distributed Systems

- ▶ Blockchain security and consensus mechanisms
- ▶ Cross-chain interoperability protocols
- ▶ Smart contract security
- ▶ Decentralized identity systems
- ▶ Hypergraph-based architectures

Ubiquitous Computing

- ▶ Context-aware computing
- ▶ Mobile and embedded systems
- ▶ IoT integration and edge computing
- ▶ Pervasive system architectures

Privacy-Preserving Computation

- ▶ Secure multiparty computation
- ▶ Zero-knowledge proofs
- ▶ Federated learning in distributed environments
- ▶ Homomorphic encryption

Cryptographic Protocols

- ▶ Protocol design for data integrity
- ▶ Privacy-preserving system architectures
- ▶ Cryptographic system verification

Technical Skills

Blockchain Technologies

- ▶ Hyperledger Fabric (Expert) - Chaincode development, network architecture, Fabric-CA identity management
- ▶ Ethereum & Solidity (Advanced) - Smart contract development, security analysis, Hardhat, OpenZeppelin
- ▶ Blockchain Interoperability (Expert) - Cross-chain atomic swaps, oracle integration, DID-based access control
- ▶ Consensus Mechanisms - Consortium blockchain design, performance benchmarking with Hyperledger Caliper

Programming Languages

- ▶ Go (Golang) - Expert - Chaincode development, microservices, gRPC integration
- ▶ Python - Advanced - Research prototyping, data analysis, MPC frameworks (MP-SPDZ)
- ▶ JavaScript/Node.js - Advanced - Mobile wallet integration, USSD gateways, REST APIs
- ▶ Solidity - Advanced - Smart contract security, Hardhat, MythX, formal verification
- ▶ Mobile Development - Android SDK, iOS, Bluetooth Low Energy (BLE), USSD/SMS integration

Cryptography & Security

- ▶ Secure Multiparty Computation (SMPC) - Expert - MP-SPDZ, threshold decryption, private set operations
- ▶ Zero-Knowledge Proofs - Expert - zk-SNARKs, Bulletproofs, R1CS, selective disclosure protocols
- ▶ Homomorphic Encryption - Advanced - Paillier cryptosystem, encrypted computation on blockchain
- ▶ Physical Unclonable Functions (PUFs) -

DevOps & Infrastructure

- ▶ Container Orchestration - Docker, Kubernetes, Helm charts, Docker Swarm
- ▶ CI/CD Pipelines - Jenkins, GitHub Actions, automated deployment for blockchain networks
- ▶ Monitoring & Observability - Prometheus, Grafana, performance dashboards, resource profiling
- ▶ Cloud Platforms - IBM Cloud, AWS IoT Core, Infura (Ethereum RPC)

identity, IoT security

- ▶ Post-Quantum Cryptography - Applied in CAESER project for future-proof security

- ▶ Microservices Architecture - Service mesh, API gateways, distributed systems design

IoT & Edge Computing

- ▶ IoT-Blockchain Integration - MQTT, sensor data anchoring, Raspberry Pi, edge gateways
- ▶ Mobile & Embedded Systems - Android SDK, BLE, GSM/GPRS modules, offline-first design
- ▶ Sensor Fusion - Multimodal data collection, real-time behavioral analytics

Research & Analysis Tools

- ▶ Blockchain Performance Benchmarking - Hyperledger Caliper, TPS analysis, latency profiling
- ▶ Smart Contract Security Analysis - MythX, Manticore, Slither, Oyente, formal verification
- ▶ Cryptographic Libraries - MP-SPDZ, HElib, ZKP libraries, Paillier implementations

References

Prof. Salil S. Kanhere

 Professor, School of Computer Science and Engineering, UNSW, Australia

- Relation: Primary PhD supervisor (2019-2024)
- Research Areas: IoT, Blockchain, Cyber-Physical Systems, Cybersecurity, Applied ML
- Contact: Available upon request

Prof. Raja Jurdak

 Professor, Director of Trusted Networks Lab, Queensland University of Technology (QUT), Australia

- Relation: PhD co-supervisor (2019-2024)
- Research Areas: IoT, Blockchain, Distributed Systems, Energy Systems
- Contact: Available upon request

Dr. Charity Wayua

 Director, IBM Research Africa

- Relation: Research manager at IBM Research Africa (2017-2019)
- Research Areas: Human-Computer Interaction, Digital Inclusion, ICT4D
- Contact: Available upon request

Dr. Sid Chi-Kin Chau

 Principal Research Scientist, CSIRO Data61, Australia

- Relation: Research manager at CSIRO Data61 (2023-2024)
- Research Areas: Cybersecurity for Energy Systems, Blockchain, IoT Security
- Contact: Available upon request